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THE HANF NUMBER OF SECOND ORDER LOGIC¹

K. JON BARWISE

Abstract. We prove, among other things, that the number mentioned above cannot be shown to exist without using some $\Pi_1(\mathcal{P})$ instance of the axiom of replacement.

§1. The existence of the Hanf number. The Hanf number of a logic L^* is the least cardinal number κ such that if a sentence of L^* has a model of cardinality $\geq \kappa$ then it has arbitrarily large models. Hanf's argument shows that one can prove in ZF (Zermelo-Fraenkel set theory with the axiom of choice) that the Hanf number of L^* exists, provided the sentences of L^* form a set.

For most logics one can replace Hanf's extremely nonconstructive proof with a "computational" proof that a certain cardinal either is the Hanf number of L^* or at least is an upper bound. Our first result shows that the nonconstructive proof is basically the only one possible for L^2 , second order logic. We show that one cannot prove that the Hanf number of L^2 exists without using some shady instance of the axiom of replacement, shady in the sense that it begs the question by using replacement for some operation F whose definition demands universal quantification over the universe of all sets. This is in contrast even to $L_{\omega_1\omega_1}$. As badly behaved as the Hanf number of $L_{\omega_1\omega_1}$ is, one can at least prove it exists without recourse to this form of replacement.

We use Z to denote the first order axioms of Zermelo set theory (extensionality, foundation, pair, union, separation, power set, infinity and the axiom of choice). It simplifies later definitions if we formulate Z with $\in$, $=$ and a unary function symbol $\mathcal{P}$ for the power set operation. Thus the power set axiom reads

$$\forall x \forall y [y \in \mathcal{P}(x) \leftrightarrow y \subseteq x]$$

(where $x \subseteq y$ is the obvious abbreviation). The theory Z suffers from serious defects, as Skolem and Fraenkel pointed out long ago. One cannot prove the existence of $\aleph_\omega$, for example, or define the sequence

$$R_\beta = \bigcup_{\alpha < \beta} \mathcal{P}(R_\alpha)$$

by recursion. One needs *some* form of the axiom of replacement, but nothing like the strength of the full schema. We study the theory ZF_1 obtained by adding to Z the replacement axioms for *definite* functions F ; definite in the sense that if $F(x) = y$ is true in some R_α then it is really true.

A quantifier of the form $\forall x(x \in t \rightarrow Q)$ or of the form $\exists x(x \in t \wedge Q)$, where t is a term of our language, is called a bounded quantifier, and is abbreviated by $\forall x \in t Q$

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or $\exists x \in tQ$. A formula in which all quantifiers are bounded is called a $\Delta_0(\mathcal{P})$ -formula. For example, if Q is a $\Delta_0(\mathcal{P})$ -formula so is $\forall x \in y \exists z \in \mathcal{P}(\mathcal{P}(x))Q$. The $\Sigma_n(\mathcal{P})$ -formulas are those of the form $(\exists x_1)(\forall x_2) \cdots (\cdots x_n)Q$ where Q is a $\Delta_0(\mathcal{P})$ -formula. The $\Pi_n(\mathcal{P})$ formulas are those of the form $(\forall x_1)(\exists x_2) \cdots (\cdots x_n)Q$ where Q is a $\Delta_0(\mathcal{P})$ -formula. A formula like $\forall x_1 \forall x_2 \exists x_3 \exists x_4 Q$ where Q is $\Delta_0(\mathcal{P})$ is equivalent in Z to a $\Pi_2(\mathcal{P})$ -formula.

We use ZF_1 to denote the first order theory whose axioms are those of Z together with all instances of $\Sigma_1(\mathcal{P})$ -replacement; that is, the universal closure of

$$\forall x \in y \exists z Q \rightarrow \exists w \forall x \in y \exists z \in w Q$$

for every $\Sigma_1(\mathcal{P})$ -formula Q in which w does not occur free. (It turns out that one gets the same theory if one requires Q to be $\Delta_0(\mathcal{P})$, or allows Q to be of the form $(q_1 x_1) \cdots (q_n x_n)S$ where S is $\Delta_0(\mathcal{P})$ and each q_i is either $\exists$ or a bounded quantifier. One also gets the same theory if one strengthens the hypothesis to $\forall x \in y \exists! z Q$.)

The theory ZF_1 is quite a strong theory, though not as strong as ZF . In ZF_1 one can define operations by recursion from old ones, provided the old ones have $\Sigma_1(\mathcal{P})$ definable graphs. For example, one can define $\aleph_\alpha$ and $\beth_\alpha$ as functions of α , the cardinality function $|x|$ = the least ordinal equinumerous with x , and prove all the usual theorems of cardinal arithmetic. One can also define the R_α 's as above and prove $\forall x \exists \alpha (x \in R_\alpha)$. The proofs of these results are exactly the proofs in ZF with a little attention paid to seeing that everything that crops up is $\Sigma_1(\mathcal{P})$. ZF_1 is a sub-theory of ZF . It seems safe to claim that ZF_1 is strong enough to prove any theorem of classical mathematics provable in ZF .

1.1. THEOREM (OF ZF_1). *If κ is the Hanf number of L^2 then R_κ is a model of ZF_1 .*

As we shall see, it is a simple matter to conclude from the proof of 1.1 that κ is in fact the κ th cardinal λ such that R_λ is a model of ZF_1 , and further Mahlo-like extensions. Given the strength of the theory ZF_1 , κ is a very large cardinal indeed.

We see from Theorem 1.1 that the existence of the Hanf number of L^2 is not provable in ZF_1 . Corollary 1.3 is a strengthening of this result.

The sentence "the Hanf number of L^2 exists" is most naturally written as a $\Sigma_3(\mathcal{P})$ -sentence. Can it be written in any simpler form, say $\Sigma_2(\mathcal{P})$ or $\Pi_3(\mathcal{P})$? A few attempts convinces one that the answer is probably no. However, since this sentence is a theorem of ZF it is equivalent, in ZF , to any other theorem like $\forall x (x = x)$, leaving one feeling cheated.

1.2. THEOREM. *Let A be any $\Pi_3(\mathcal{P})$ -sentence. If*

$$ZF_1 \vdash (A \rightarrow \exists \kappa [\kappa \text{ is the Hanf number of } L^2])$$

then $ZF \vdash \neg A$.

The sentence in question is thus not equivalent, in ZF_1 , to any $\Pi_3(\mathcal{P})$ -sentence since the only time a $\Pi_3(\mathcal{P})$ -sentence A can imply it in ZF_1 is when A is already refutable in ZF . In connection with 1.2 note that the axioms "there are arbitrarily large inaccessible (measurable) cardinals" and "there are arbitrarily large strongly compact cardinals" are $\Pi_2(\mathcal{P})$ and $\Pi_3(\mathcal{P})$ respectively.

1.3. COROLLARY. *Let T be an extension of ZF_1 which is consistent with ZF and is obtained by adding $\Pi_3(\mathcal{P})$ -sentences to ZF_1 . Then the existence of the Hanf number of L^2 is not a theorem of T .*

PROOF. Apply Theorem 1.2, the compactness theorem and the fact that the conjunction of a finite number of $\Pi_n(\mathcal{P})$ -sentences is equivalent in ZF_1 to another $\Pi_n(\mathcal{P})$ -sentence. $\dashv$

We now turn to the proofs of Theorems 1.1 and 1.2. As the reader will have noticed, we are using capital letters P, Q, S to denote formulas of the language of set theory. We assume that the development of the syntax and semantics of first and second order logic has been carried out in a reasonable way in the theory ZF_1 . (This is implicit in the statement of Theorem 1.1.) We use φ, ψ , etc. as variables ranging over first order formulas of the formalized first order language of set theory; Φ, Ψ , etc. as variables ranging over second order formulas of the formalized second order language. To every real formula Q of the language of set theory there corresponds its formalization or "Gödel number," which we think of as denoted by an eliminable term $\ulcorner Q \urcorner$ of the language of set theory.

Let us denote the formula " κ is the Hanf number of L^2 " by $H(\kappa)$.

1.4. LEMMA (OF $\text{ZF}_1 + H(\kappa)$). *If a sentence Φ of L^2 has, for each $\lambda < \kappa$, a model of power $\geq \lambda$ then Φ has arbitrarily large models.*

PROOF. It suffices to consider the case where Φ mentions only equality since we can quantify out any other relation, function or constant symbols. Given Φ , consider the sentence $\exists < \Psi$ where Ψ is the conjunction of

" $<$ is a well-ordering"

$\forall x \exists y [x < y \wedge \text{"}\Phi \text{ holds relativized to } \{z: z < y\}\text{"}]$.

If all models of Φ have power $< \kappa$ but Φ has a model of power $\geq \lambda$ for each $\lambda < \kappa$ then $\exists < \Psi$ has a model of power κ and hence has arbitrarily large models, in which case Φ also has arbitrarily large models. $\dashv$

We use $\mathcal{R}_\beta$ to denote the structure $\langle R_\beta, \varepsilon \upharpoonright R_\beta, \mathcal{P} \upharpoonright R_\beta \rangle$. A cardinal λ is *uniformly κ -definable* if there is a first order formula $\theta(x)$ with only x free such that, for all $\beta \geq \kappa$,

$\lambda = \text{the unique } a \in R_\beta \text{ such that } \mathcal{R}_\beta \models \theta[a]$.

1.5. LEMMA (OF $\text{ZF}_1 + H(\kappa)$). *For every $\alpha < \kappa$ there is a uniformly κ -definable cardinal λ with $\alpha < \lambda < \kappa$.*

PROOF. By Lemma 1.4, κ is a limit cardinal. Choose a sentence Φ of L^2 which has a model of power $> |\alpha|$ but none of power $\geq \kappa$. Let λ be the least cardinal such that Φ has no model of power $\geq \lambda$ and let $\theta(\lambda)$ express this definition in the natural way. Then $\lambda < \kappa$ by 1.4 and $\theta(x)$ is a uniform κ -definition of λ . (We are assuming here that second order logic has been formalized so that every sentence Φ of L^2 is in R_ω and hence can be described inside $\theta(x)$.) $\dashv$

1.6. LEMMA (OF $\text{ZF}_1 + H(\kappa)$). *If a first order sentence φ is true in $\mathcal{R}_\kappa$ then φ is true in $\mathcal{R}_\beta$ for arbitrarily large β .*

PROOF. To prove 1.6 consider the sentence $(\varphi \wedge \Psi)$ where Ψ is a sentence of L^2 whose models are just those structures isomorphic to some $\mathcal{R}_\beta$. $\dashv$

PROOF OF 1.1. Let us work in $\text{ZF}_1 + H(\kappa)$. Actually we show that if κ is any cardinal satisfying Lemmas 1.5 and 1.6, then $\mathcal{R}_\kappa$ is a model of (formalized) ZF_1 . By 1.6, $\kappa > \omega$. For any limit ordinal $\beta > \omega$, $\mathcal{R}_\beta \models Z$ so we need only prove that $\mathcal{R}_\kappa$ is a model of $\Sigma_1(\mathcal{P})$ -replacement. Let $\psi(x, y, z)$ be a $\Sigma_1(\mathcal{P})$ formula and suppose that,

for certain $a, b \in R_\kappa$,

$$\mathcal{R}_\kappa \models \forall x \in a \exists y \psi(x, y, b)$$

but that

$$\mathcal{R}_\kappa \models \neg \exists w \forall x \in a \exists y \in w \psi(x, y, b).$$

(The case where ψ has more free variables $z_1 \cdots z_n$ follows from this case, using ordered pairs.) Let $\alpha < \kappa$ be such that $a, b \in R_\alpha$ and use 1.5 to choose a λ , $\alpha < \lambda < \kappa$, and a uniform κ -definition $\theta(x)$ of λ . The following sentence is true in $\mathcal{R}_\kappa$ and hence by 1.6, in $\mathcal{R}_\beta$ for arbitrary large $\beta \geq \kappa$:

$$\exists z [\theta(z) \wedge \exists u \in R_z \exists v \in R_z [\forall x \in u \exists y \psi(x, y, v) \wedge \neg \exists w \forall x \in u \exists y \in w \psi(x, y, v)]]].$$

Thus, for arbitrarily large $\beta \geq \kappa$ there are $a, b \in R_\lambda$ such that $Q(\beta, a, b)$, where $Q(\beta, a, b)$ is the predicate,

$$\mathcal{R}_\beta \models [\forall x \in a \exists y \psi(x, y, b) \wedge \neg \exists w \forall x \in a \exists y \in w \psi(x, y, b)].$$

Q is equivalent to a $\Sigma_1(\mathcal{P})$ predicate since the relation " $a_1 \cdots a_n$ satisfies $\varphi(v_1 \cdots v_n)$ " in $\mathcal{R}_\beta$ " is a $\Delta_1(\mathcal{P})$ -relation of β , $a_1 \cdots a_n$, φ . Using the axiom of separation define

$$X = \{ \langle a, b \rangle \in R_\lambda \times R_\lambda : Q(\beta, a, b) \text{ for some } \beta \}.$$

Now for all $\langle a, b \rangle \in X$ there is a least β such that $Q(\beta, a, b)$ so, by $\Sigma_1(\mathcal{P})$ -replacement, there is a γ greater than all these least β 's. But $\exists a \exists b Q(\beta, a, b)$ holds for arbitrarily large β 's, so for some $\beta_1 > \gamma$ and, some $a, b \in R_\lambda$, $Q(\beta_1, a, b)$. But then $Q(\beta_0, a, b)$ holds for some $\beta_0 < \gamma < \beta_1$, by the definition of γ .

Thus $\mathcal{R}_{\beta_0} \models \forall x \in a \exists y \psi(x, y, b)$ and hence $\mathcal{R}_{\beta_1} \models \forall x \in a \exists y \in R_{\beta_0} \psi(x, y, b)$ since $\mathcal{R}_{\beta_0} \models \psi(x, y, b)$ and $\psi \in \Sigma_1(\mathcal{P})$ implies $\mathcal{R}_{\beta_1} \models \psi(x, y, b)$. But this contradicts $Q(\beta_1, a, b)$ by taking $w = R_{\beta_0}$. $\downarrow$

Before turning to the proof of Theorem 1.2 let us show (in ZF_1) that κ is indeed the κ th cardinal λ such that $\mathcal{R}_\lambda$ is a model of ZF_1 .

First note that if $\mathcal{R}_\alpha \models \text{ZF}_1$ then α is a cardinal (since otherwise we could code α up as a relation $r \subseteq \beta \times \beta$ for some $\beta < \alpha$, but then $r \in R_\alpha$ and hence its order type α would be in R_α). Next, notice that it follows from 1.1 and 1.6 that if $\lambda < \kappa$ is uniformly κ -definable then there is an α , $\lambda < \alpha < \kappa$, such that $\mathcal{R}_\alpha \models \text{ZF}_1$. From 1.5 it follows that the sequence of $\mathcal{R}_\alpha$, for $\alpha < \kappa$, which are models of ZF_1 has order type λ for some limit ordinal $\lambda \leq \kappa$. Assume $\lambda < \kappa$ and define $f(\beta) =$ the β th ordinal α such that $\mathcal{R}_\beta \models \text{ZF}_1$, for $\beta < \lambda$; $f(\beta) = 0$ for $\beta \geq \lambda$. We can define f in $\mathcal{R}_\kappa$ by a $\Sigma_1(\mathcal{P})$ formula with parameter λ and by $\Sigma_1(\mathcal{P})$ -replacement $\kappa = \bigcup_{\alpha < \lambda} f(\alpha) \in R_\kappa$, a contradiction. Thus $\lambda = \kappa$. $\downarrow$

We now turn to the proof of Theorem 1.2. Let $\text{True}_3(x)$ be the natural $\Sigma_3(\mathcal{P})$ -formula with one free variable x which defines truth for $\Sigma_3(\mathcal{P})$ -sentences. In particular, for each $\Sigma_3(\mathcal{P})$ -sentence Q , we can prove (in ZF_1) $\text{True}_3(\ulcorner Q \urcorner) \leftrightarrow Q$. The reader familiar with Lévy [3] will have no trouble giving the actual definition, tedious though it may be.

1.7. LEMMA (OF $\text{ZF} + H(\kappa)$). *For every $\Sigma_3(\mathcal{P})$ -sentence φ , if $\mathcal{R}_\kappa \models \varphi$ then $\text{True}_3(\varphi)$.*

PROOF. The proof is very much like the proof of Theorem 1.1, that $\mathcal{R}_\kappa \models \Sigma_1(\mathcal{P})$ -replacement, but it is easier since we are working in ZF . Using 1.5 and 1.6 one shows that if $\mathcal{R}_\kappa \models \exists x \forall y \exists z \varphi(x, y, z)$, where $\psi \in \Delta_0(\mathcal{P})$ then there is a $\lambda < \kappa$ such that for arbitrarily large β there is an $a \in R_\lambda$ such that $\mathcal{R}_\beta \models \forall y \exists z \psi(a, y, z)$. Using the axiom of replacement (we are working in ZF now, not ZF_1) one gets a single $a \in R_\lambda$ such

that the above holds for arbitrarily large β . But then $\forall y \exists z \psi(a, y, z)$ is true, so $\text{True}_3(\exists x \forall y \exists z \psi)$. $\dashv$

PROOF OF 1.2. Let A be the $\Pi_3(\mathcal{P})$ -sentence $\forall x \exists y \forall z B(x, y, z)$ where B is $\Delta_0(\mathcal{P})$ and suppose that $\text{ZF}_1 + A \vdash \exists \kappa H(\kappa)$. Let us work in ZF and prove $\text{True}_3(\ulcorner \exists x \forall y \exists z \neg B(x, y, z) \urcorner)$, and hence $\neg A$. Let φ be $\ulcorner \exists x \forall y \exists z \neg B(x, y, z) \urcorner$ and ψ be $\ulcorner \exists \kappa H(\kappa) \urcorner$. Thus $\text{ZF}_1 \vdash (\neg \varphi \rightarrow \psi)$. Let κ be the Hanf number of L^2 . By Theorem 1.1., $\mathcal{R}_\kappa \models \text{ZF}_1$ but by Lemma 1.4, $\mathcal{R}_\kappa \models \neg \psi$ so $\mathcal{R}_\kappa \models \varphi$. Hence, by Lemma 1.7, $\text{True}_3(\varphi)$. $\dashv$

§2. An application. The downward Löwenheim-Skolem Theorem for a logic L^* usually gives one a cardinal valued bounding function F such that if a sentence of L^* has a model of power $> \lambda$ then it has one of some power μ with $\lambda \leq \mu \leq F(\lambda)$. If L^* is first order logic or $L_{\omega_1 \omega}$ then $F(\lambda) = \lambda + \aleph_0$. If L^* is $L_{\omega_1 \omega_1}$ then $F(\lambda) = (\lambda + \aleph_0)^{\aleph_0}$, this in spite of the fact that the Hanf number of $L_{\omega_1 \omega_1}$ is very badly behaved. In this section we show that any $\Sigma_1(\mathcal{P})$ -function F which one might hope to use as a bounding function for L^2 fails for arbitrarily large λ 's. Thus any meaningful downward Löwenheim-Skolem Theorem for L^2 is refutable in ZF. (It is trivial to give a useless $\Pi_1(\mathcal{P})$ bounding function for L^2 .) The proof is rather amusing since it uses Theorem 1.2. In order to give the reader some appreciation for our result we list below some $\Sigma_1(\mathcal{P})$ -definable functions on cardinals.

- (i) $F(\lambda) = \aleph_0 + \aleph_\lambda$,
- (ii) $F(\lambda) = \lambda$,
- (iii) $F(\lambda) = \aleph_\lambda$,
- (iv) $F(\lambda) = \beth_\lambda = |R_\lambda|$,
- (v) $F(\lambda) = G(H(\lambda))$ (where G and H are also $\Sigma_1(\mathcal{P})$ functions from cardinals to cardinals).
- (vi) $F(\lambda) =$ the λ th cardinal κ such that $G(\kappa) = \kappa$ (where G is a $\Sigma_1(\mathcal{P})$ function on cardinals which is continuous and increasing, like (i)–(iv)).
- (vii) $F(\lambda) =$ the λ th inaccessible (measurable) cardinal (assuming that there are arbitrarily large such cardinals).

2.1. THEOREM. *For each $\Sigma_1(\mathcal{P})$ -formula $Q(x, y)$ with only x and y free the following is a theorem of ZF: Assume that for every cardinal λ there is a unique cardinal μ such that $Q(\lambda, \mu)$ and write $F(\lambda) = \mu$ if $Q(\lambda, \mu)$. There is a sentence Φ of L^2 such that for arbitrarily large λ , Φ has a model $\mathfrak{A}$ of power $|\mathfrak{A}| > \lambda$ but none with $\lambda \leq |\mathfrak{A}| \leq F(\lambda)$.*

PROOF. If we can show in ZF that for each λ there is a sentence Φ of L^2 and a $\lambda' \geq \lambda$ such that Φ has a model $\mathfrak{A}$ with $|\mathfrak{A}| > \lambda'$ but none with $\lambda' \leq |\mathfrak{A}| \leq F(\lambda')$ then we can use the axiom of replacement to get one Φ which works for arbitrarily large λ . But the statement in italics is equivalent to a $\Pi_2(\mathcal{P})$ -sentence, hence a $\Sigma_3(\mathcal{P})$ -sentence, so to prove it we need only assume its negation and prove, within ZF_1 , that the Hanf number of L^2 exists; then apply Theorem 1.2. Its negation implies that for some λ , a sentence Φ of L^2 has a model $\mathfrak{A}$ with $|\mathfrak{A}| \geq \lambda' \geq \lambda$ iff it has one with $\lambda' \leq |\mathfrak{A}| \leq F(\lambda')$. If X is the set of sentences of L^2 which do not have arbitrarily large models, then $\Phi \in X$ iff there is a $\lambda' \geq \lambda$ such that Φ has no models $\mathfrak{A}$ with $\lambda' \leq |\mathfrak{A}| \leq F(\lambda')$. For each $\Phi \in X$ let $g(\Phi)$ be the least cardinal $\lambda' \geq \lambda$ such that Φ has no models $\mathfrak{A}$ with $\lambda' \leq |\mathfrak{A}| \leq F(\lambda')$, and hence none with $|\mathfrak{A}| \geq \lambda'$.

The function g is $\Sigma_1(\mathcal{P})$ definable (using parameters X and λ) so $\kappa = \sup\{g(\Phi) : \Phi \in X\}$ exists by $\Sigma_1(\mathcal{P})$ -replacement. If Φ has a model of power κ then it has arbitrarily large models. Thus some $\kappa_0 \leq \kappa$ is the Hanf number of L^2 . $\dashv$

This proof shows, by the way, how one can prove in ZF_1 that the Hanf number of $L_{\omega_1\omega_1}$ exists.

§3. Concluding remarks. The notion of “definite”, as used by Zermelo [5] in his original formulation of Z was quite vague. Skolem’s suggestion in [4] that one use “first order property” for “definite property” explicitly presupposes that one is willing to form the universe V of all sets,

$$(3.1) \quad V = \bigcup_{\alpha} R_{\alpha}$$

(the union is taken over all ordinals) and consider it as a meaningful mathematical totality. In fact, this was one of Skolem’s arguments against using set theory as a foundation for mathematics. Another possibility, however, is to deny any meaning to (3.1) and use a more restricted notion of definite, something like $\Delta_0(\mathcal{P})$ -property.

If one takes this seriously, one might insist that only $\Delta_0(\mathcal{P})$ -formulas be allowed in the axiom of separation, as well as in the axiom of replacement (for replacement $\Delta_0(\mathcal{P})$ and $\Sigma_1(\mathcal{P})$ amount to the same thing, but not for separation). Call the resulting theory ZF_0 . One can still do a lot of set theory in ZF_0 . For example, all of the explicit claims made for ZF_1 prior to Theorem 1.1 hold for ZF_0 . **The first trouble one encounters is that one cannot prove**

$$(3.2) \quad \text{Every well ordering is isomorphic to an ordinal.}$$

This uses a Π_1 or Σ_1 instance of the axiom of separation. In fact, it is consistent with ZF_0 to assume

$$(3.3) \quad \forall x \exists \alpha [x \in R_{\alpha} \text{ and } \alpha \text{ is isomorphic to a recursive well ordering of the integers}].$$

(One can see this by using a theorem of Friedman (cf. §6 of Barwise [1]) and the fact that the standard part of a model of ZF_0 is again a model of ZF_0 .) As a consequence, a lot of interesting set theory and model theory cannot be carried out in ZF_0 . For example, in $ZF_0 + (3.3)$ one can show that there is no Hanf number for ω -logic, hence none for $L_{\omega_1\omega}$ or $L_{\omega_1\omega_1}$. On the other hand, the usual Hanf number computations for ω -logic and $L_{\omega_1\omega}$ can be carried out in $ZF_0 + (3.2)$, hence in ZF_1 .

This paper has its origins in Friedman’s underground polemic [2], in particular in §2 where he raises questions concerning the axiom of replacement. He formulates a theory T on p. 9 and asks whether one can prove the existence of the Hanf numbers of $L_{\omega_1\omega_1}$ or L^2 in T . The reader can easily check that T is equivalent to ZF_0 so that the answer is negative by the above remarks. If one goes from ZF_0 all the way to ZF_1 then the Hanf number of $L_{\omega_1\omega_1}$ exists, but still not that of L^2 . We do not know whether or not one can show in $ZF_0 + (3.2)$ that the Hanf number of $L_{\omega_1\omega_1}$ exists.

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